

**MASTER OF COMPUTER
APPLICATIONS (NEW)
(MCA-NEW)**

**Term-End Examination
December, 2021**

**MCS-221 : DATA WAREHOUSING
AND DATA MINING**

Time : 3 hours

Maximum Marks : 100

(Weightage : 70%)

Note : *Question no. 1 is **compulsory**. Answer any **three** questions from the rest.*

1. (a) Define a Data Warehouse. List and explain the four characteristics of a Data Warehouse. 10
- (b) Explain the following OLAP architectures and draw their architectural diagram : 10
 - (i) Multidimensional Online Analytical Processing (MOLAP)
 - (ii) Hybrid Online Analytical Processing (HOLAP)

- (c) Define “data cleaning” which is a data preprocessing technique. In this context, explain the concept of Noisy data cleaning along with some suitable examples. 10
- (d) What is a Decision Tree ? How is it useful in classification ? With the help of an example, explain the process of construction of a decision tree and its representation. 10
2. (a) What is Text Mining ? Where is it used ? Explain any two text mining techniques with the help of a suitable example for each. 10
- (b) In the context of mining multimedia data on the web, explain the following terms : 10
- (i) Page Rank
 - (ii) Hits
 - (iii) Page Layout Analysis
 - (iv) Vision Page Segmentation
3. (a) Define a Data Lake. Explain the step-by-step process of creating a data lake. 10
- (b) With the help of an example, explain the star schema dimensional model. 10

4. (a) What is Metadata ? What are its contents ?
Justify how metadata can be an important component in Data Warehousing ? Also mention its types. 10

(b) Explain the three components of ETL. Also mention how to improve the ETL performance. 10

5. Write short notes on any **four** of the following : 4×5=20

(a) Data Marts

(b) Association Rule Generation

(c) Apriori Algorithm

(d) Clustering and its Methods

(e) Bayes' Theorem

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Term-End Examination

June, 2022

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Note : (i) *Question No. 1 is compulsory.*

(ii) *Attempt any **three** questions from the rest.*

1. (a) What is Association Rule Mining ? With reference to this, explain the concepts given below :

10

- (i) Support Count

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- (ii) Frequent Item set
 - (iii) Association Rule
 - (iv) Rule Evaluation metrics
- (b) Define classification. With the help of an example for each, explain the following classification models : 10
- (i) Descriptive modeling
 - (ii) Predictive modeling
- (c) Describe the snowflake schema multidimensional modeling technique. Give an example and illustrate it. List its pros and cons. 10
- (d) What is complex data modeling ? Where is this used ? Explain “Anchor” complex data model. 10
2. (a) List and describe all the four types of OLAP operations. 10
- (b) Define text mining. Mention any *three* applications of it. What are the various techniques used to analyze the web-usage patterns ? 10

3. (a) Define a data warehouse. List and explain all the 3 types of data warehouses. 10
- (b) What is data transformation ? List and explain all the data transformation steps. 10
4. (a) What is data mining ? List out the key differences between data warehousing and data mining. Mention any *four* applications of data mining. 10
- (b) What is ETL ? Why do you need ETL in data warehouses ? Also, explain how ETL works in Data Warehousing ? 10
5. Write short notes on any *four* of the following : 4×5=20
- (a) OLTP
- (b) Data Marts
- (c) Web structure mining
- (d) Cloud Data Warehousing
- (e) Data Granularity

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1. (a) Define Dimensional Modelling. With reference to this, define the terms Facts, Fact Table, Dimensions and Dimensional Table. Give an example use-case and derive fact table and dimension tables. 10
- (b) With the help of a Data Warehouse Architecture diagram, explain the following components and their significance : 10
 - (i) ETL
 - (ii) Metadata
 - (iii) Data Warehouse Access Tools
 - (iv) Data Warehouse Reporting Layer

- (c) Define a Decision Tree. With the help of an example, explain the construction and representation of decision tree. Also, mention its strengths and weaknesses. 10
- (d) Discuss the following categories of Data Mining Issues : 10
- (i) Mining Methodology and User Iteration Issues
 - (ii) Performance-based Issues
 - (iii) Diverse Data Types Issues
2. (a) Define Data Cleaning. Explain the ways and means of handling the Missing Values and Noisy Data while data preprocessing. 10
- (b) Write and explain the K-means algorithm for clustering. How does it work ? 10
3. (a) Why does dimensionality reduction of text need to be done ? Explain Tokenization process and Vector from text approach, with the help of a suitable example for each. 10
- (b) Enumerate the best practices for Data Warehouse Architecture. 5
- (c) Describe the following types of data marts : 5
- (i) Dependent data marts
 - (ii) Independent data marts

4. (a) With the help of an example use-case, explain the Snowflake schema. List its advantages and disadvantages. 10
- (b) Define Web Mining. What kind of tasks can be performed using it ? Discuss its features and also a few applications. 10
5. Write short notes on the following : 4×5=20
- (a) Data Transformation (with reference to data preprocessing)
- (b) Cloud Data Warehousing
- (c) Data Lake and its Architecture
- (d) Data Warehouse Automation
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Note : (i) *Question No. 1 is compulsory.*

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rest.*

1. (a) Discuss the following data preprocessing stages briefly : 10

- (i) Data Cleaning
- (ii) Data Integration
- (iii) Data Reduction
- (iv) Data Transformation

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- (b) Define Aggregate fact tables and derived dimension tables. What are their significance ? Give an example. Also mention their advantages and disadvantages. 10
- (c) Enumerate the key challenges in data warehouse design. 5
- (d) Differentiate between a data lake and a data warehouse. 5
- (e) Write and explain the Apriori algorithm to identify the most frequently occurring elements and meaningful associations in a dataset. 10
2. (a) What is Cluster Analysis ? How is this used in Data Mining ? Give an example. Also mention few applications of cluster analysis in data mining. 10
- (b) List and discuss various types of Web-mining. 10
3. (a) Explain the following techniques for Dimensionality Reduction : 10
- (i) Feature Selection
- (ii) Feature Extraction
- (b) Discuss the layered implementation of ETL in a Data Warehouse. 10

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4. (a) Define OLAP. Differentiate between Multicube and Hypercube. Mention the applications of OLAP reporting system. 10
- (b) List and explain the following types of Data Warehouses : 10
- (i) Enterprise Data Warehouse (EDW)
- (ii) Operational Data Store
5. Write short notes on the following : 4×5=20
- (a) ELT *vs.* ETL
- (b) Data Marts
- (c) Applications of Data Mining
- (d) OLAP data cube operations

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Note : (i) *Question No. 1 is compulsory.*

(ii) *Answer any **three** questions from the rest.*

1. (a) With the help of a diagram, describe the Conceptual Architecture of Hadoop Data Warehouse. 10
- (b) Draw and explain star schema diagram and snow-flake schema diagram for the dimensions (Products, Customers, Time, Locations) and fact (Sales-Items) for the measures namely Quantity-sold and Amount-sold for a manufacturing company data warehouse dimensional modeling. 10

- (c) Define Noisy data while doing data pre-processing. Delete the noise with Binning smoothing techniques for the following details using partition in Bins (Equal-frequency) :

4, 2, 6, 10, 8, 16, 12, 24, 22, 14, 26

stored price details (in dollars). 10

- (d) Define Clustering in Data Mining. Write and explain k-means clustering algorithm. List its advantages and disadvantages. 10

2. (a) What is Web-Mining ? List various web-mining tasks. Also, discuss the following types of web-mining : 10

(i) Web content mining

(ii) Web usage mining

- (b) With the help of an example, explain rule-based classification. 5

- (c) What are the various steps involved in building a classification model ? Explain with the help of an example. 5

3. (a) With the help of an example, explain Market Basket Analysis. 5
- (b) Write and explain Apriori algorithm used to identify the most frequently occurring elements and meaningful associations in any dataset. 10
- (c) List and discuss any *two* popular data mining tools. 5
4. (a) Discuss ETL and its need. Explain in detail, all the steps involved in ETL with the help of a suitable diagram. 10
- (b) List and explain any *three* key challenges of Data Warehouse. 3
- (c) With reference to Alex Gorelik, explain the following additional data lake stages : 7
- (i) Data Puddle
 - (ii) Data Pond
 - (iii) Data Lake
 - (iv) Data Ocean

5. Write short notes on the following : $4 \times 5 = 20$

- (a) Aggregate fact table and derived dimensional tables
- (b) Data swamp
- (c) Data Preprocessing stages
- (d) Agglomerative approach of Hierarchical method

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Note : *Question number-1 is compulsory. Attempt any three questions from the rest.*

1. (a) With the help of a diagram, explain the real-time data warehouse architecture. Also, discuss its trade-off. 10
- (b) Describe fact constellation scheme. How is it different from snow-flake scheme ? With the help of an example explain fact constellation schema. Mention its advantages and limitations. 10

P. T. O.

- (c) Write and explain K-NN classification algorithm in Data-Mining. Discuss its advantages and disadvantages. 10
- (d) Discuss vector space modeling for representing text documents. With reference to this modeling, explain TF-IDF and Inverse Document Frequency (IDF). 10
2. (a) Discuss various factors to be considered to improve the performance of ETL. 7
- (b) Define a Data Mart. How is it different from a centralized data warehouse ? Discuss the structure of Data Mart with the help of an example. 8
- (c) Discuss the following OLAP data cube operations : 5
- (i) Roll-up (ii) Drill-down.
3. (a) Discuss the following Data Integration issues : 10
- (i) Schema Integration and object modeling
- (ii) Redundancy

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(iii) Detection and Resolution of Data Value Conflicts.

- (b) Write and explain Decision Tree Classifier algorithm with the help of an example. 10
4. (a) Discuss Linear Discriminant Analysis (LDA) and Principal Component Analysis (PCA) feature extraction techniques. 10
- (b) Explain Association Rule Mining (ARM). Also, explain the two phases involved in ARM process with the help of an example. 10
5. Write short notes on the following : $4 \times 5 = 20$
- (a) ELT and its need
- (b) Metadata and Data Warehousing
- (c) Rule based classification
- (d) Naive-bayes Classifier